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Aim of the experiment :- Design a 3X8 decoder and implement a boolean function using decoder.

Objective

(i) To design and implemnt 3X8 decoder

(ii) To Implement full adder using 3X8 decoder

Apparatus Required

Sl NO	Name of the component	Specification	Quantity
1.	Trainer kit (analog and digital)	XPO-ANADIG01	1
2.	Bread board	—	1
3.	OR Gate (IC-74LS32)	Quad 2input OR Gate	2
4.	Decoder	IC74LS138	1
5.	Connecting wires	—	As per Required

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3×8 decoder

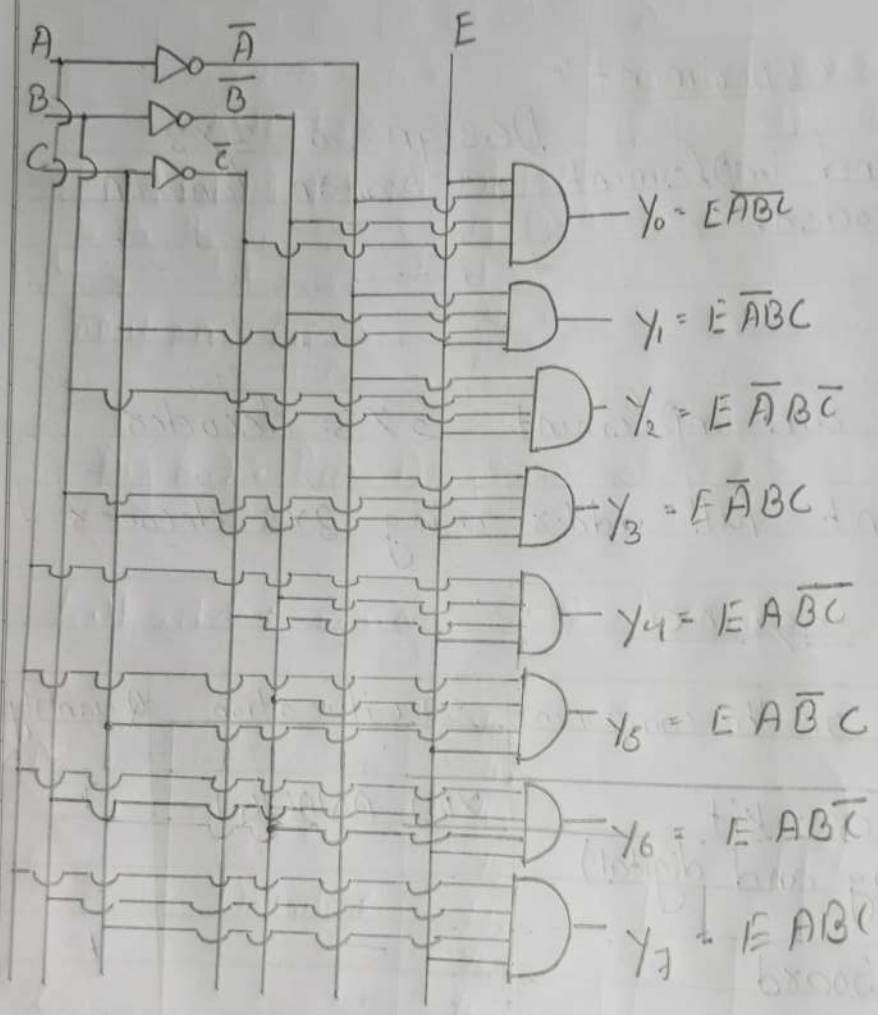


Fig 6.1:- Logic circuit diagram of 3×8 decoder

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Theory:-

(i) 3 x 8 decoder

A 3 x 8 decoder is a combinational logic circuit having 3 input lines and 8 output lines. It converts a 3-bit binary input into one of the 8 possible out lines. For each combination of inputs, only one output becomes High (logic 1), while all other output remain Low (logic 0). The inputs are represented by A, B, C the output are represented by Y_0 to Y_7.

Truth Table:-

Table 6.1:-

Enable	Input			Output							
(E)	A	B	C	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	X	X	X	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	1

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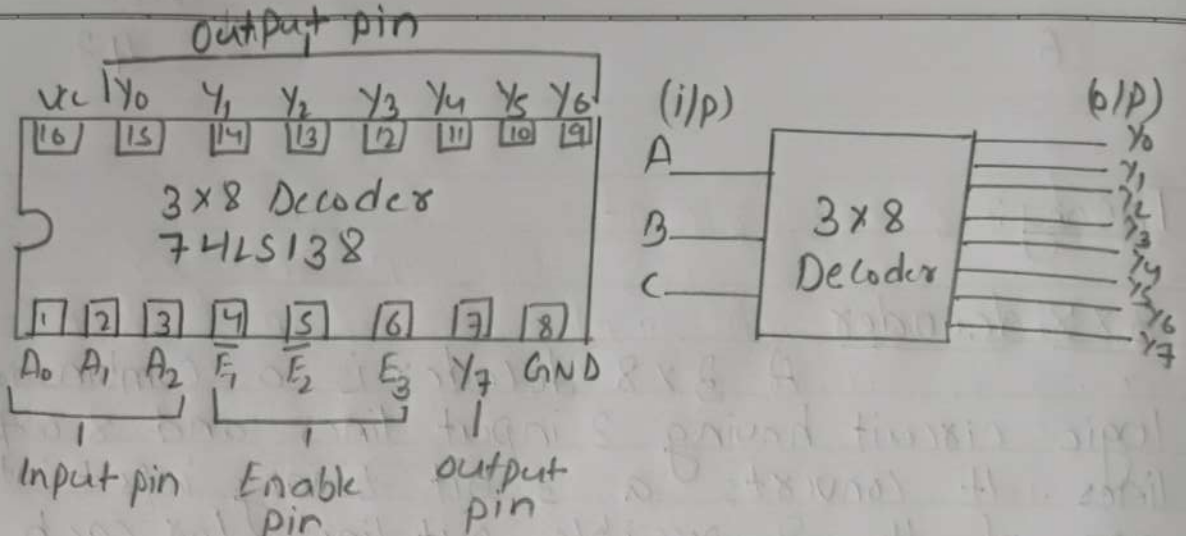


Fig 6.2: Pin Diagram of 3×8 Decoder (IC: 74LS138)

Full adder using 3×8 decoder

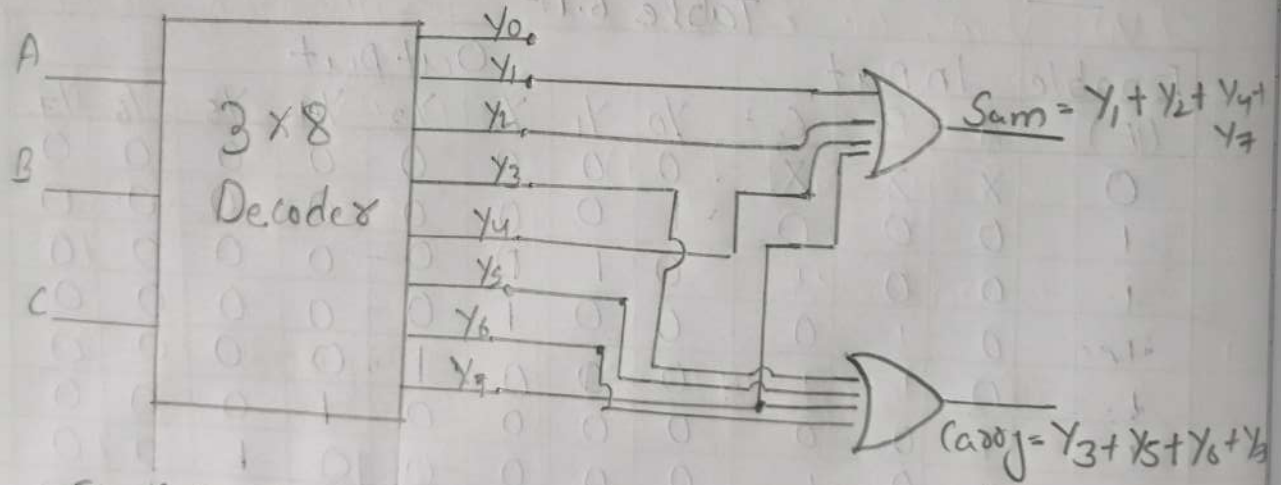


Fig 6.3: Logic circuit diagram of full adder using 3×8 decoder

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Boolean expression :-

$Y_0 = E\bar{A}\bar{B}\bar{C}$ $Y_4 = EA\bar{B}\bar{C}$

$Y_1 = E\bar{A}\bar{B}C$ $Y_5 = EA\bar{B}C$

$Y_2 = E\bar{A}B\bar{C}$ $Y_6 = EAB\bar{C}$

$Y_3 = E\bar{A}BC$ $Y_7 = EABC$

(ii) Full Adder using 3x8 decoder

A full adder is a combinational logic circuit that adds three binary inputs, namely A, B, and input(Cin) and produce two output: Sum (S) and carry (Cout)

A 3x8 decoder is used to implement the full adder by generating all eight minterms of the three input variable. The decoder output are combined using OR gates to obtain the Sum and carry outputs.

Boolean Expressions

The Sum out is High for minterm 1, 2, 4, and 7

$S = \Sigma m(1,2,4,7)$

The Carry output is High for minterms 3, 5, 6, and 7

$Cout = \Sigma m(3,5,6,7)$

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Truth Table :-

Table 6.2 :-

Input			Output	
A	B	C	Sum (S)	Carry (Co)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Procedure

A. Design of 3X8 Decoder

1. Connecting the three input lines A, B and C to the input terminals of the decoder.
2. Connecting the enable line E to logic High to enable the decoder
3. Connecting the eight outline Y_0 to Y_7 to the respective output terminal
4. Applying all possible combinations of binary inputs to the decoder.

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5. Observing the output for each input combination and recording the result in the truth table.

6. Verifying the outputs by comparing them with the theoretical truth table.

B Implementing of Full adder using 3x8 decoder

1. Connecting the inputs A, B, and Cin to the three input lines of 3x8 decoder.

2. Connecting the enable line E to logic HIGH to enable the decoder

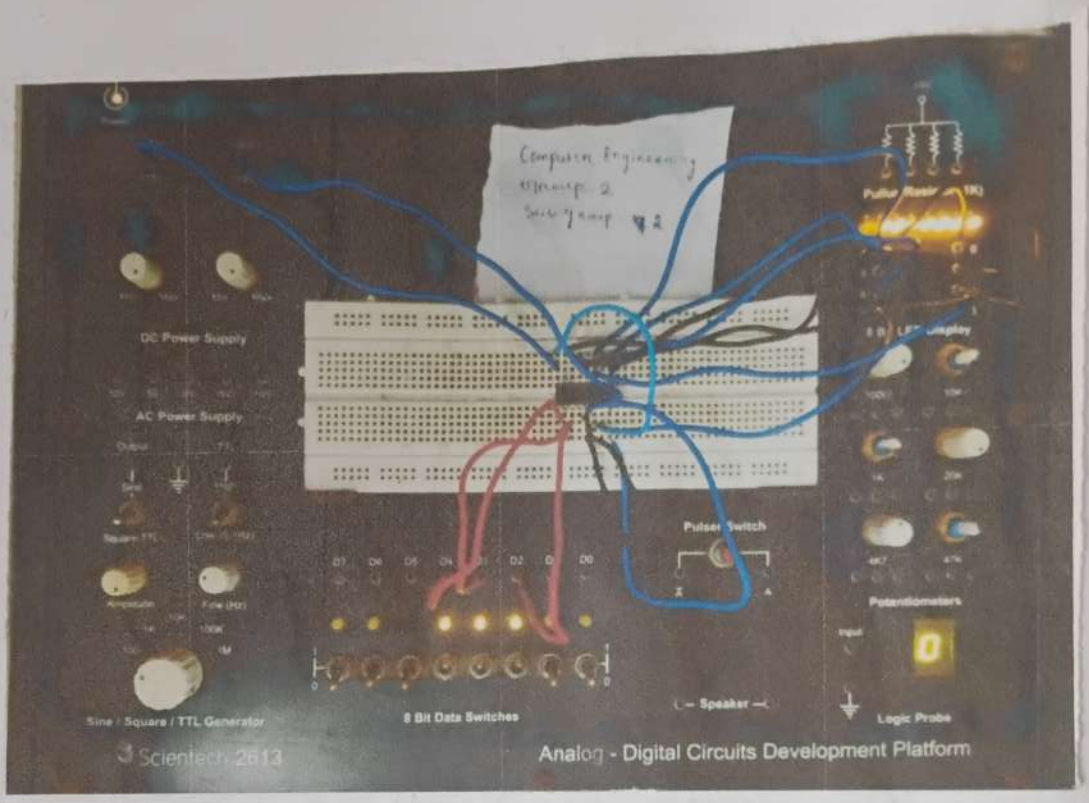
3. Connecting the decoder outputs Y_1, Y_2, Y_4 and Y_7 to a 4-input OR gate to obtain the Sum output

4. Connecting the decoder outputs Y_3, Y_5, Y_6 and Y_7 to another 4-input OR gate to obtain the carry output

5. Applying all possible combinations of binary input A, B, and Cin to the decoder.

6. Observing and recording the sum and Carry output for each input combination.

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Enable	Input			Output							
(E)	A (D_4)	B (D_3)	C (D_2)	Y_0 (0)	Y_1 (1)	Y_2 (2)	Y_3 (3)	Y_4 (4)	Y_5 (5)	Y_6 (6)	Y_7 (7)
OFF	ON	ON	ON	G	G	G	G	G	G	G	NG

(ON=1, OFF=0) (NG=Not Glowing G=Glowing)

Fig 6.4: Practical Implementation of 3x8 Decoder

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7. Verifing the obtained outputs by comparing them with the theoretical truth table of the full adder.

Observation Table

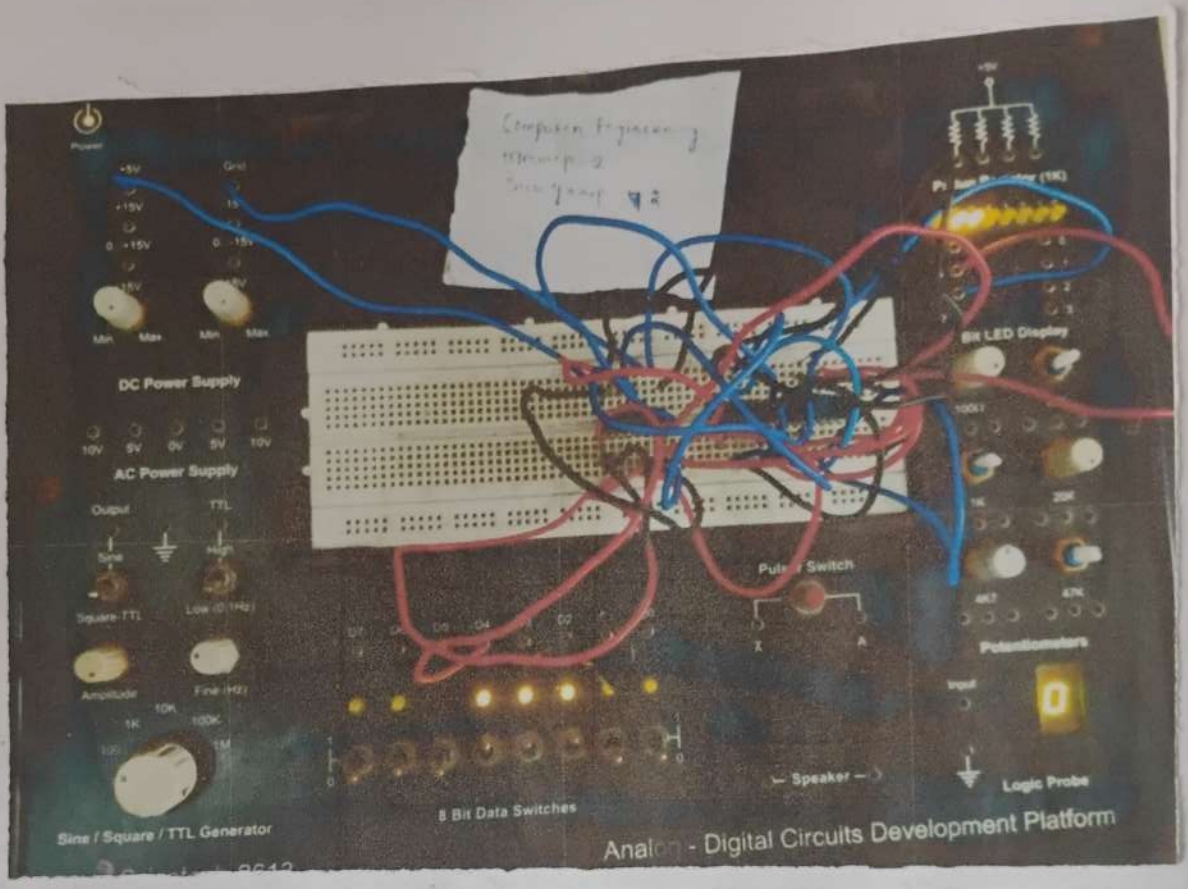
A. Observation Table for 3x8 decoder

Observation Table 1:-

Enable (E)	Input A	B	C	Output Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	Remark
ON	X	X	X	G	G	G	G	G	G	G	G	
OFF	OFF	OFF	OFF	NG	G	G	G	G	G	G	G	The practical
OFF	OFF	OFF	ON	G	NG	G	G	G	G	G	G	output
OFF	OFF	ON	OFF	G	G	NG	G	G	G	G	G	matches
OFF	OFF	ON	ON	G	G	G	NG	G	G	G	G	the theor-
OFF	ON	OFF	OFF	G	G	G	G	NG	G	G	G	tical
OFF	ON	OFF	ON	G	G	G	G	G	NG	G	G	output
OFF	ON	ON	OFF	G	G	G	G	G	G	NG	G	
OFF	ON	ON	ON	G	G	G	G	G	G	G	NG	

(ON=1, OFF=0) (NG- Not Glowing, G - Glowing)

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Input			Output	
A(D_4)	B(D_3)	C(D_2)	Sum(S)	Carry(C)
ON	ON	ON	Glowing	Glowing

(ON=1, OFF=0)

Fig 6.5: Pratical Implementation of Full adder using 3×8 decoder

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B Observation Table for Adder using 3X8 decoder

Observation Table 2:-

Input			Output		
A	B	C	Sum (S)	Carry (Co)	Remark
(D_4)	(D_3)	D_2	(6)	(5)	
OFF	OFF	OFF	NG	NG	
OFF	OFF	ON	G	NG	
OFF	ON	OFF	G	NG	The pratical output matches the theoritical output
OFF	ON	ON	NG	G	
ON	OFF	OFF	G	NG	
ON	OFF	ON	NG	G	
ON	ON	OFF	NG	G	
ON	ON	ON	G	G	

(ON = 1, OFF = 0) (NG- Not Glowing, G- Glowing)

Conclusion

The 3X8 decoder has been successfully designed and its outputs have been verified for all eight input combinations. The full adder has been implemented using the decoder and OR gates. The Sum and Carry outputs have been obtained and verified according to the full-adder truth table. Thus the correct operation of both circuits has been demonstrated

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